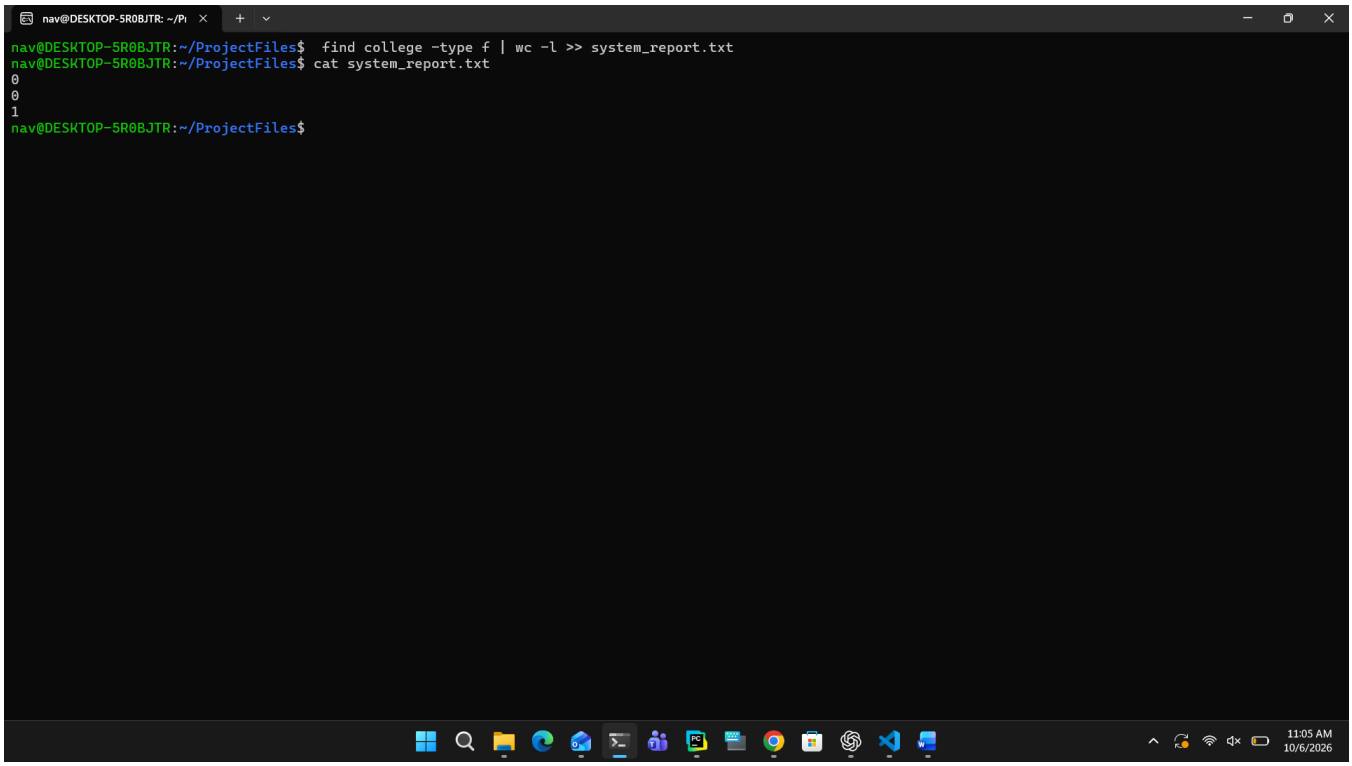
Question 7

Generate a system/directory report using find, du, df, stat, wc, sort, head, tail, uname, hostname, history, free, ps, top with screenshots and summary.

Step 1 – Count files in the College hierarchy

Commands used: find College -type f | wc -l >> system_report.txt.

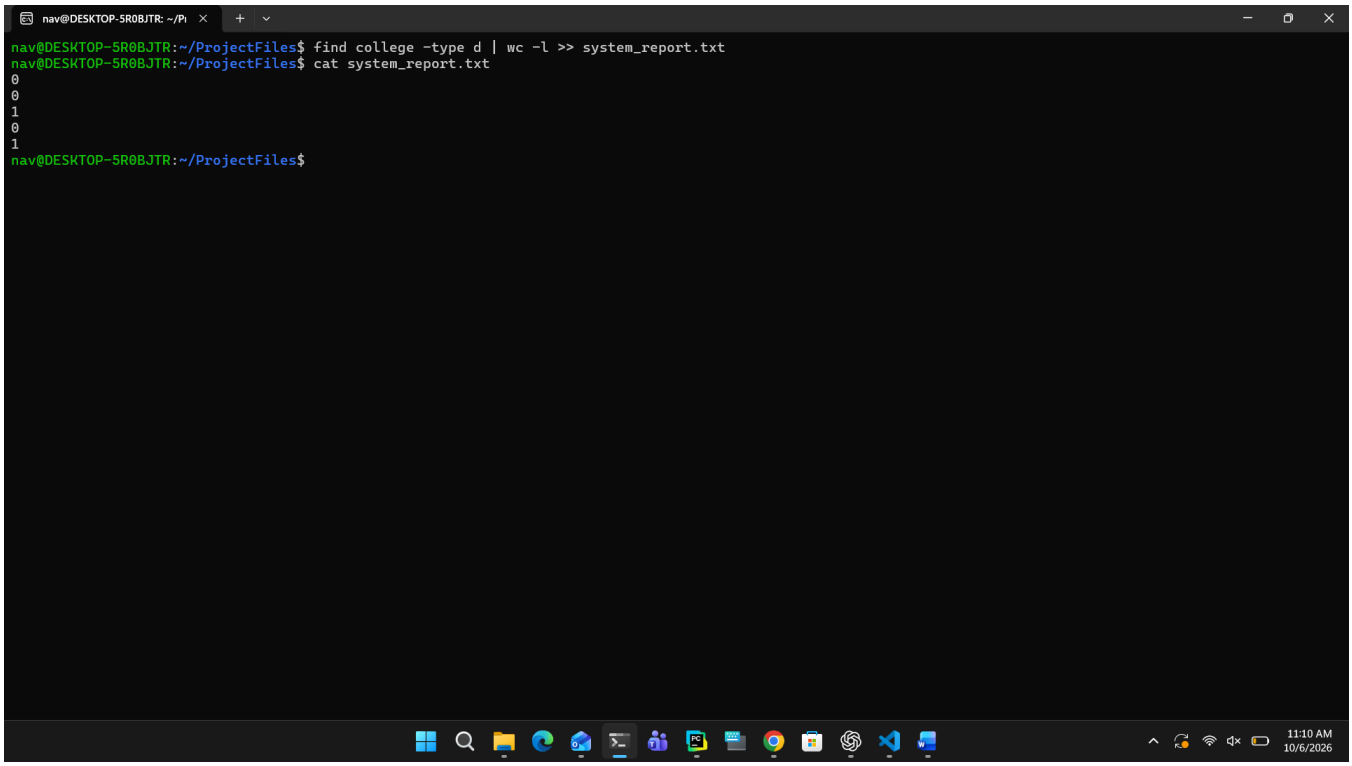
We counted all files inside the College hierarchy and appended the resulting count to system_report.txt.



Step 2 – Count directories in the College hierarchy

Commands used: find College -type d | wc -l >> system_report.txt.

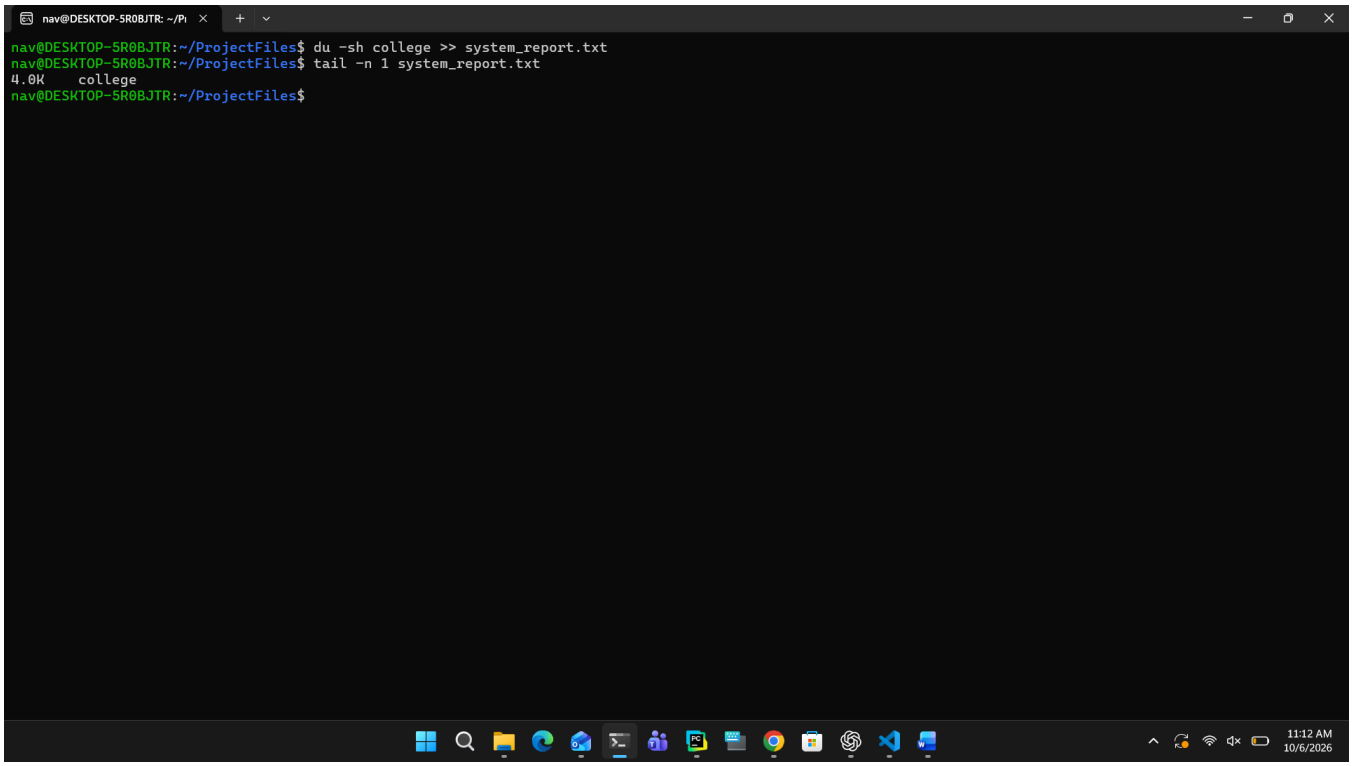
We counted all directories inside the College hierarchy and appended the resulting count to system_report.txt.



Step 3 – Check College directory disk usage

Commands used: du -sh College >> system_report.txt.

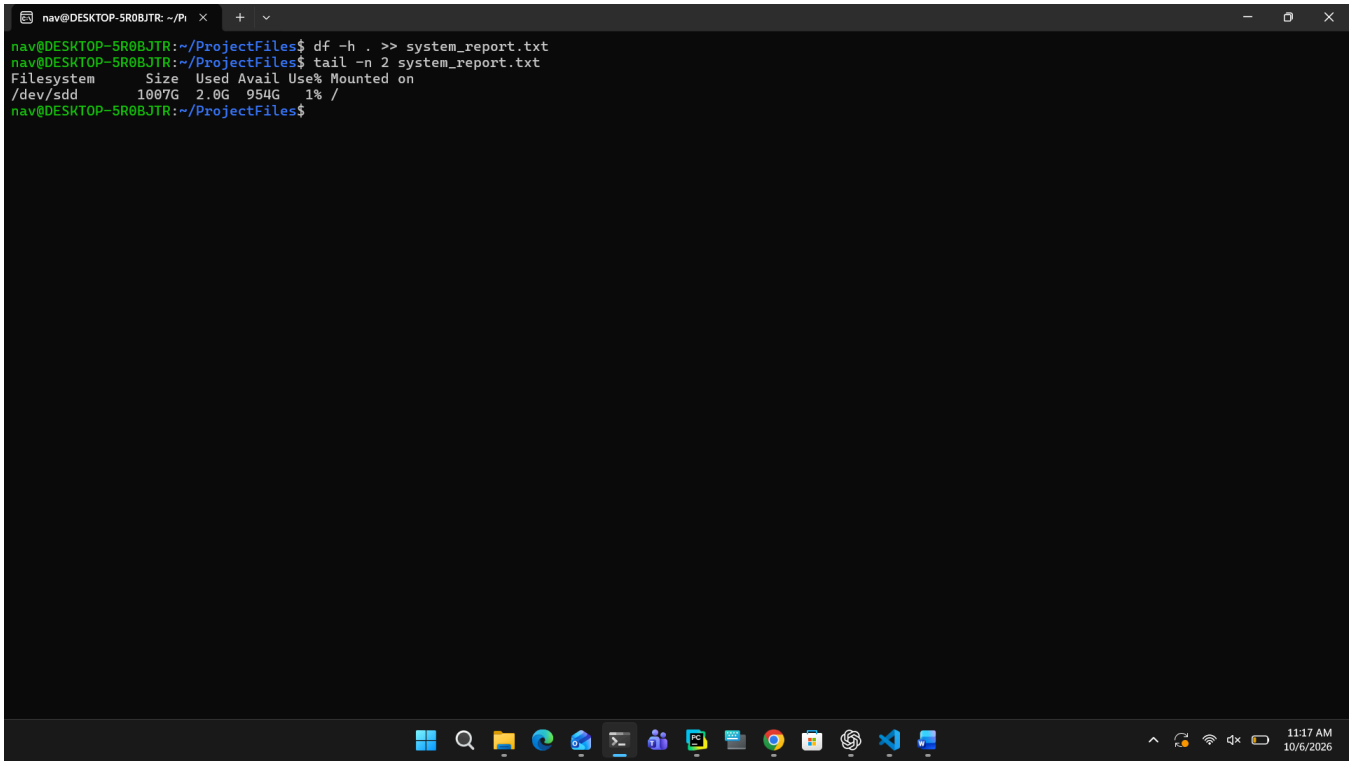
We calculated the total human-readable disk usage of the College directory and appended the result to the report.



Step 4 – Check filesystem capacity

Commands used: df -h . >> system_report.txt.

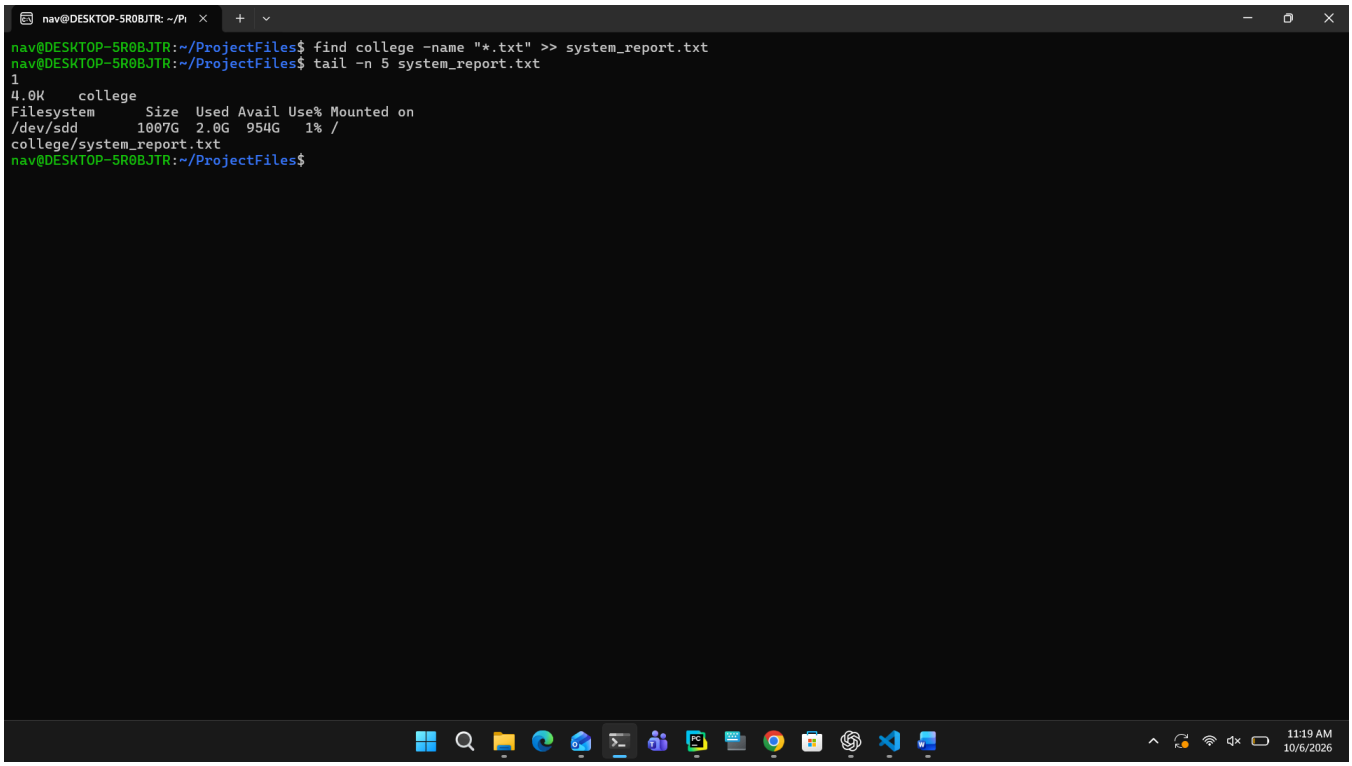
We displayed the filesystem capacity for the current mount in human-readable form and appended the result to the report.



Step 5 – Find text files

Commands used: find College -name "*.txt" >> system_report.txt.

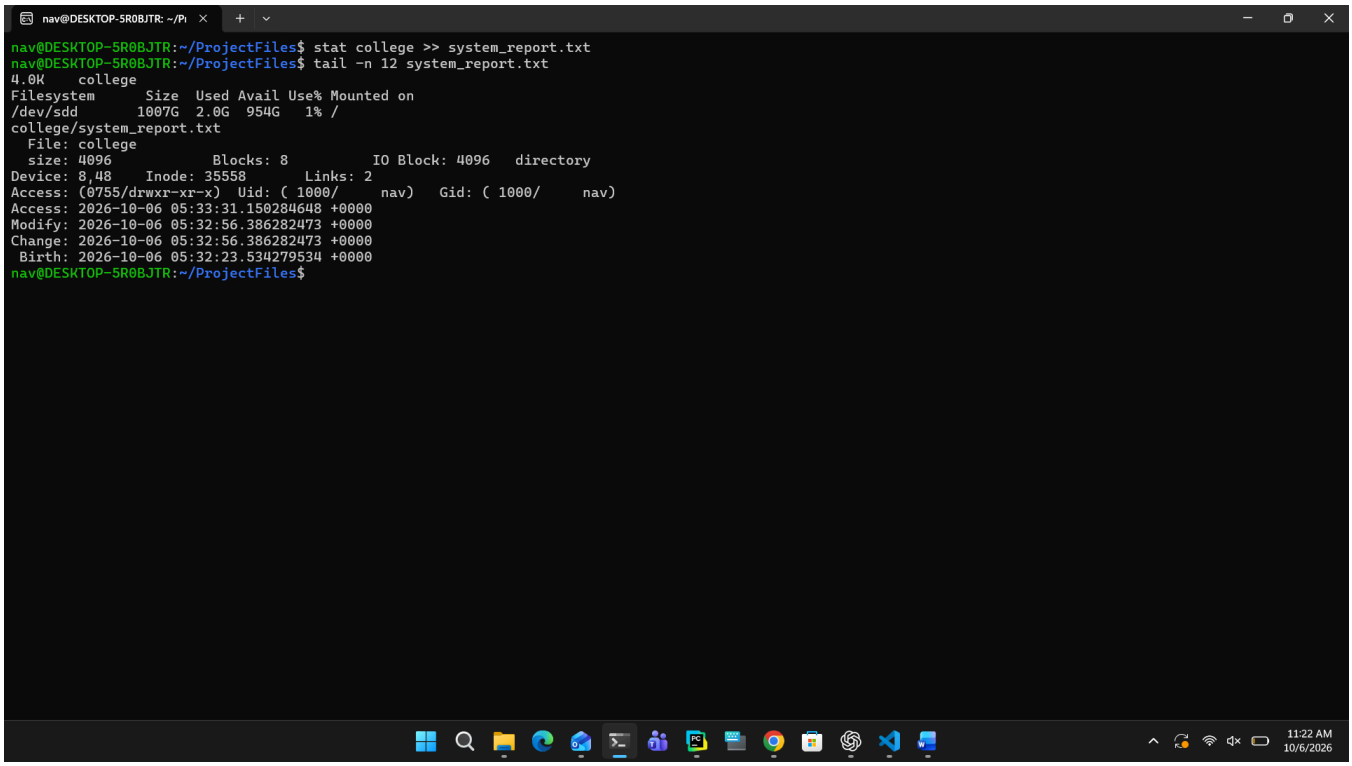
We searched the College hierarchy for .txt files and appended the matching paths to the report.



Step 6 – Display College metadata

Commands used: stat College >> system_report.txt.

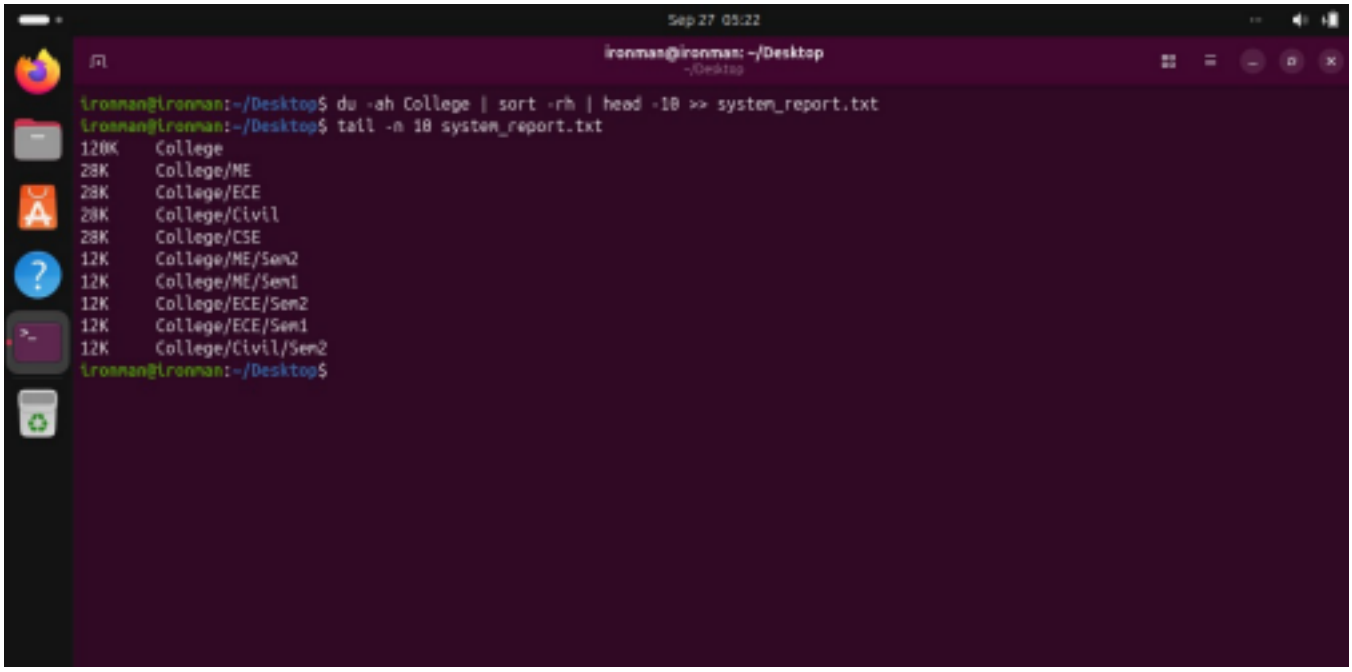
We collected metadata for the top-level College directory, including permissions, timestamps, size and inode information.



Step 7 – Rank the largest College entries

Commands used: du -ah College | sort -rh | head -10 >> system_report.txt.

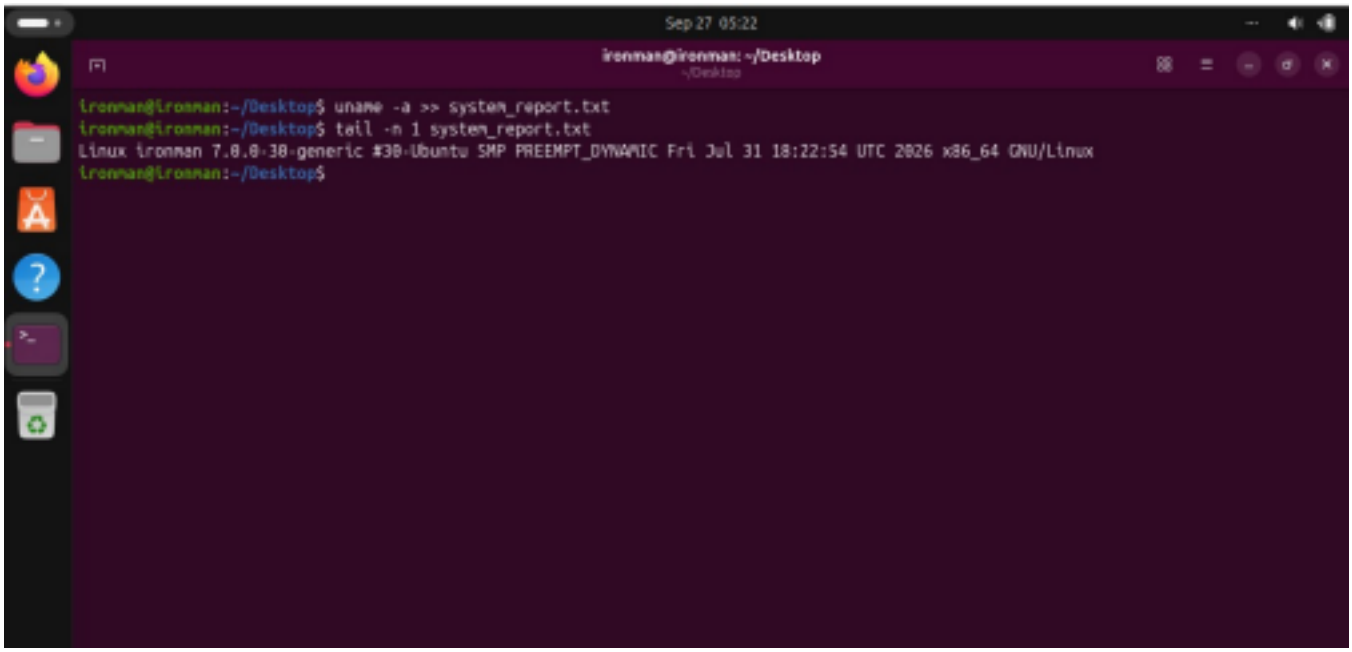
We calculated sizes for entries in College, sorted them in reverse human-readable order and recorded the ten largest entries.



Step 8 – Collect system information

Commands used: uname -a >> system_report.txt.

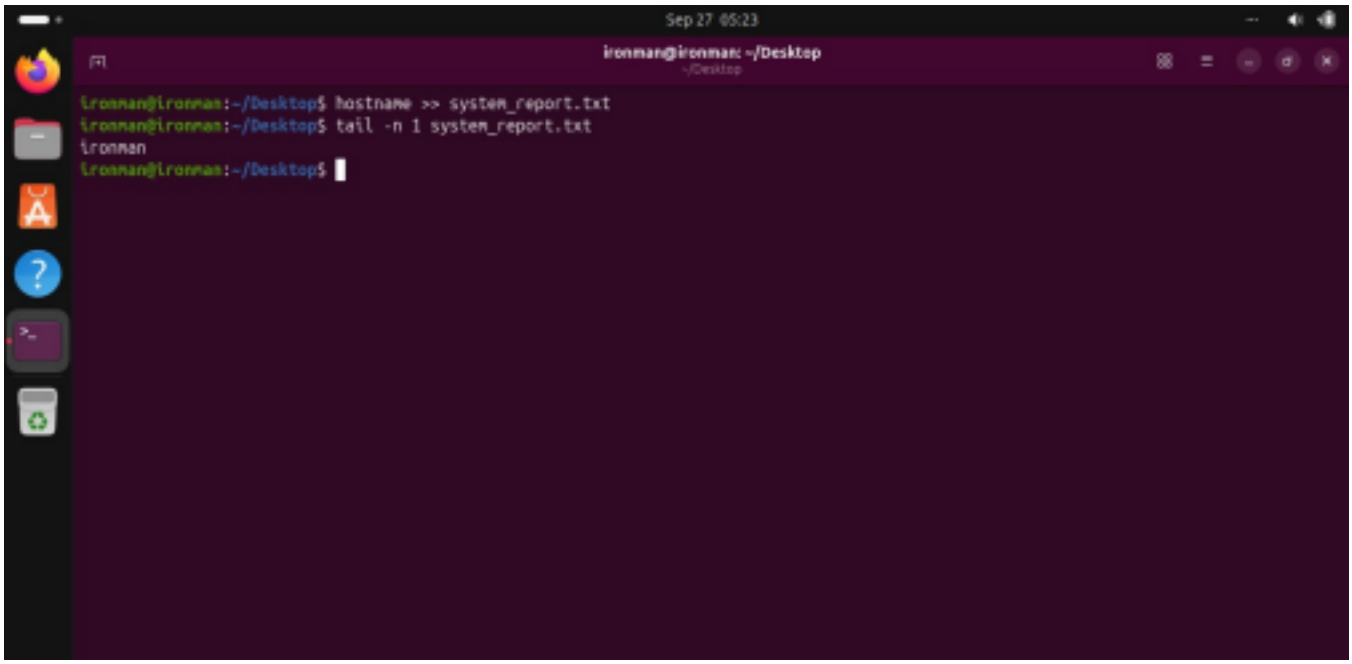
We recorded kernel, operating-system and architecture information in the report.



Step 9 – Display the hostname

Commands used: hostname >> system_report.txt.

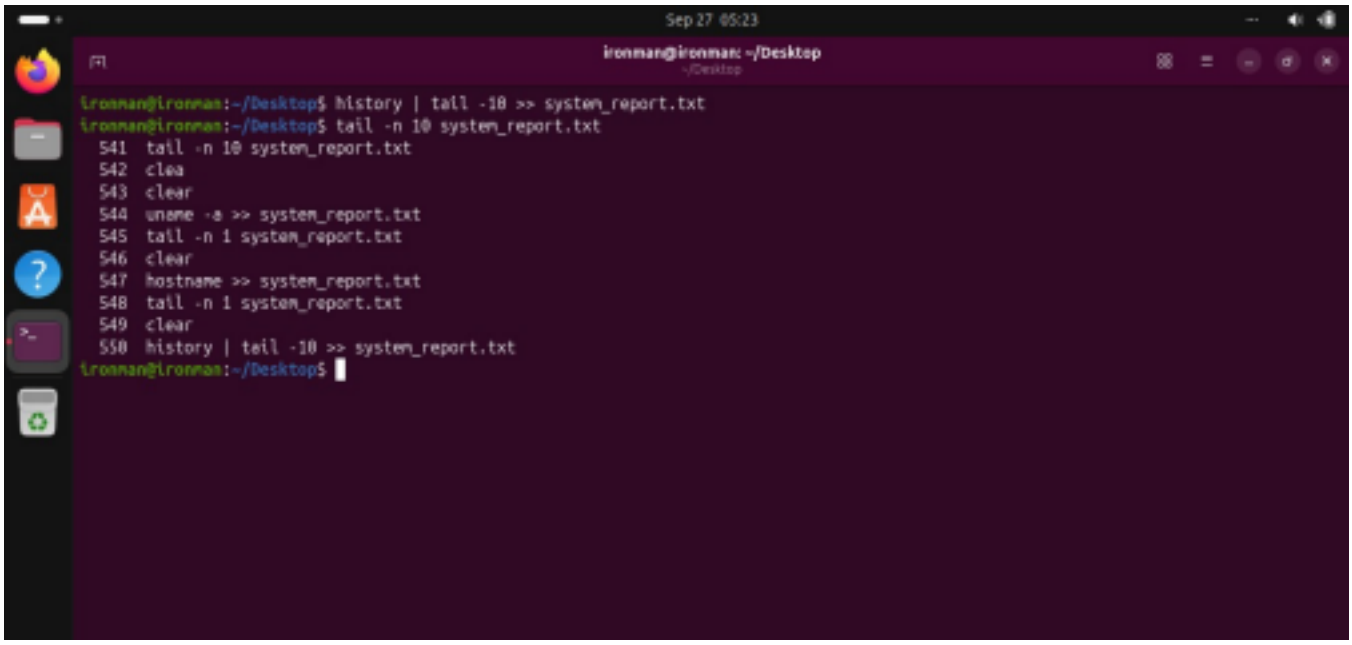
We recorded the machine hostname in the consolidated report.



Step 10 – Record recent shell history

Commands used: history | tail -10 >> system_report.txt.

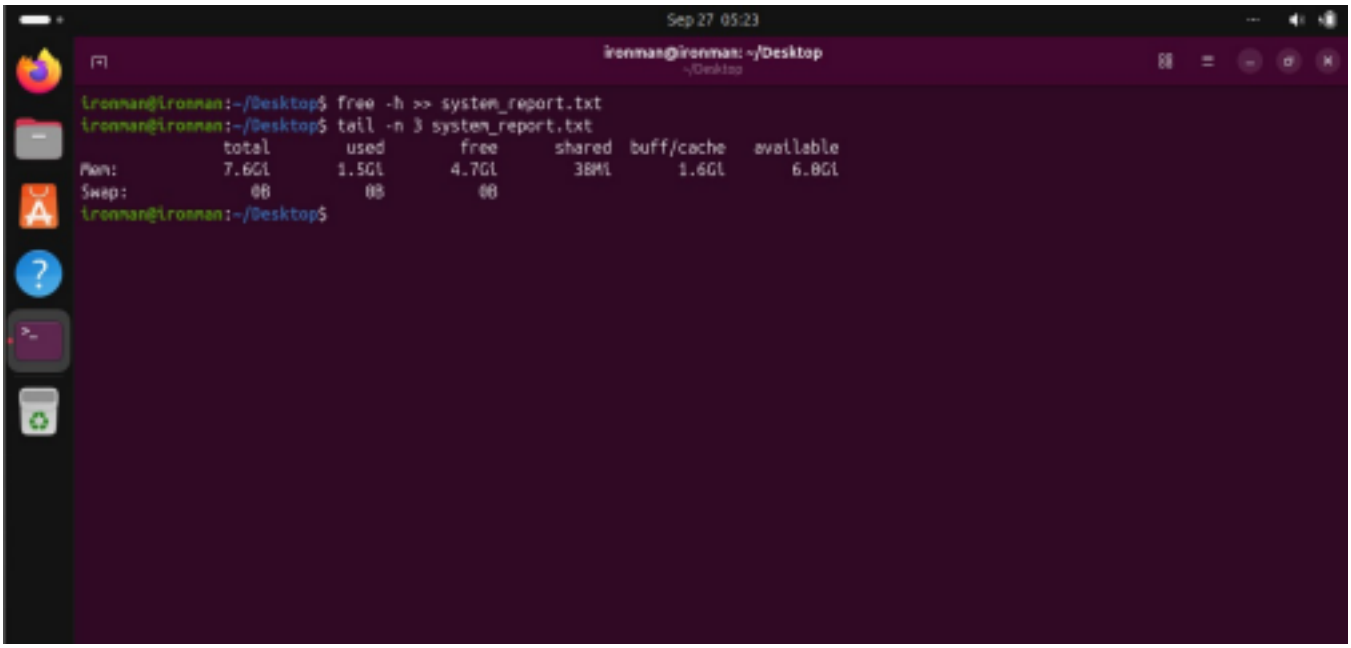
We appended the ten most recent shell commands from the current session to the report.



Step 11 – Check memory and swap usage

Commands used: free -h >> system_report.txt.

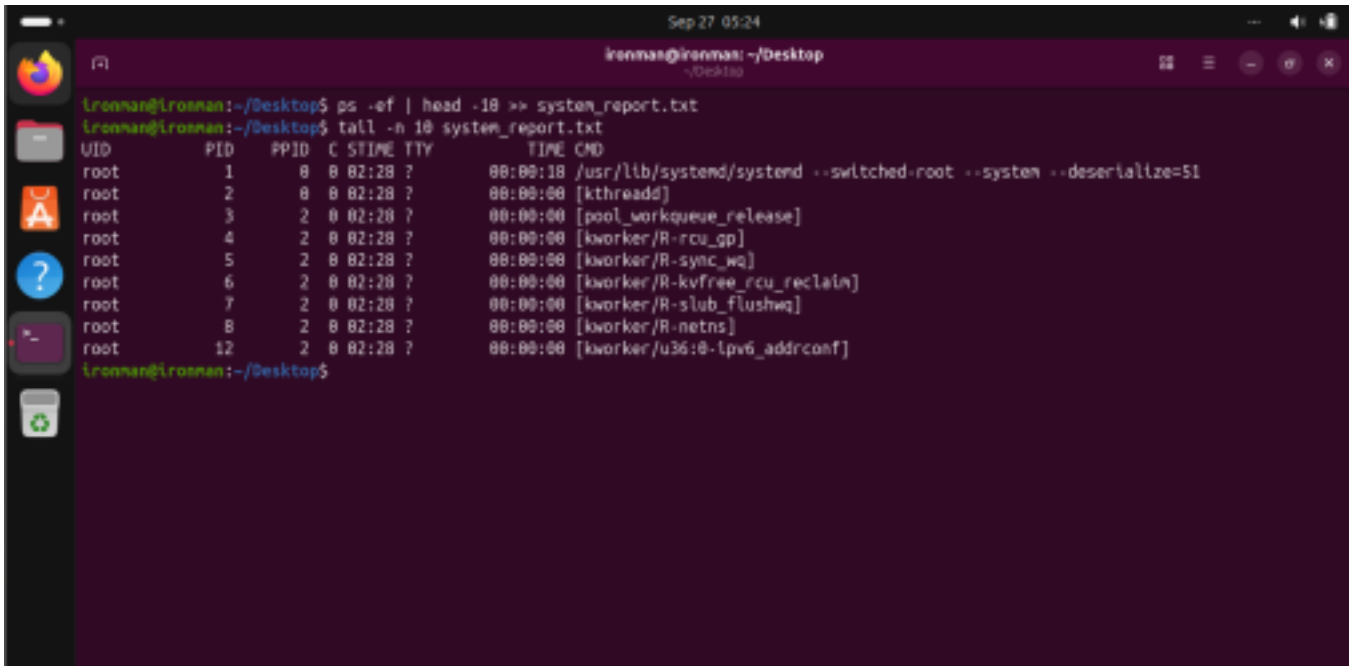
We recorded current memory and swap usage in human-readable form.



Step 12 – Display running processes

Commands used: ps -ef | head -10 >> system_report.txt.

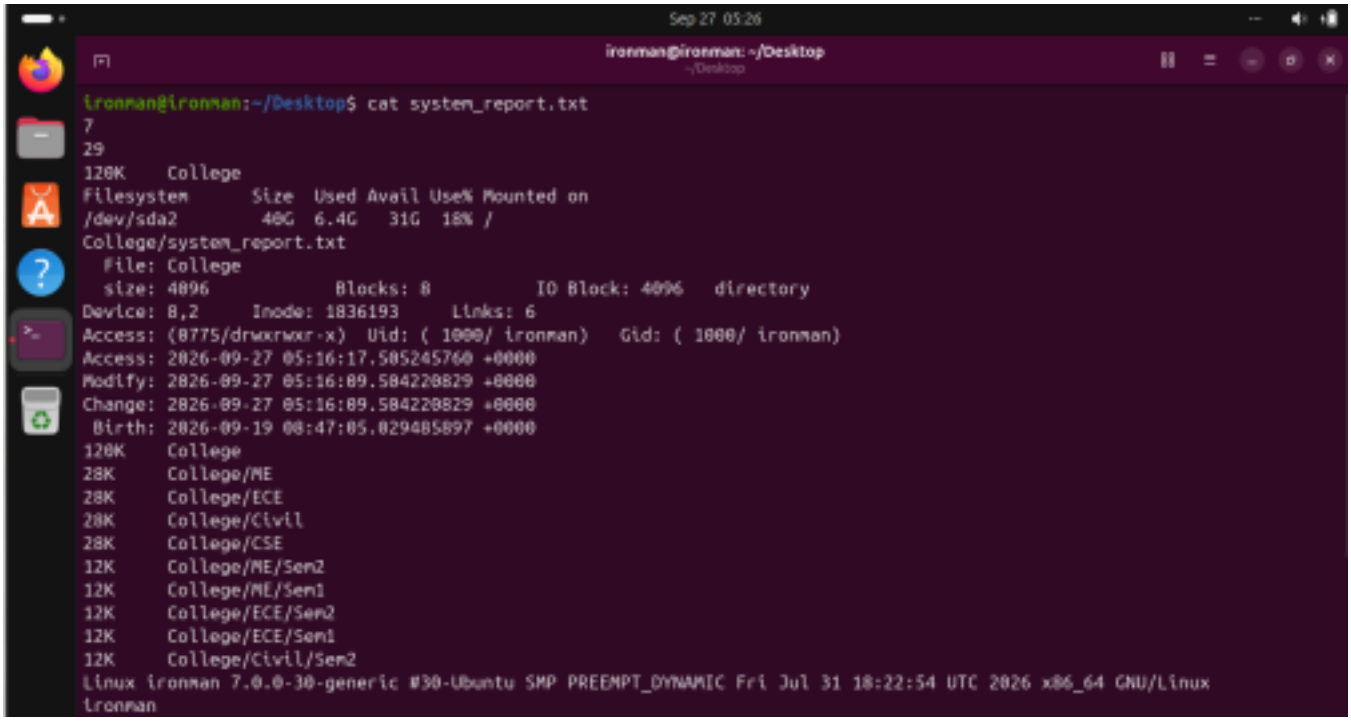
We recorded a snapshot of the first ten running processes in the report.



Step 13 – Display the completed system report

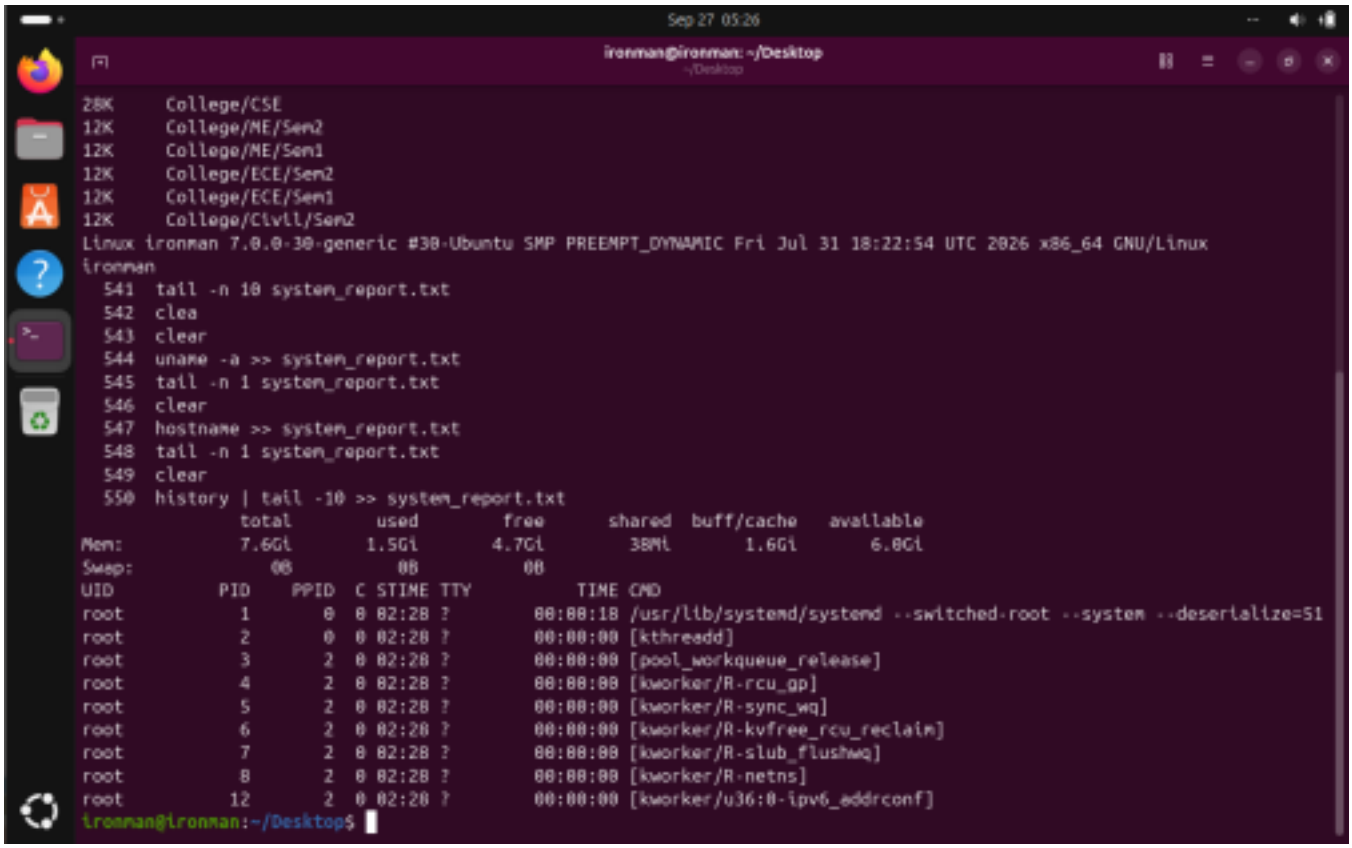
Commands used: cat system_report.txt.

We displayed the completed consolidated report. The output was longer than one terminal screen, so the final evidence is shown across two screenshots.



Step 13 – Display the completed system report (continued)

Continuation of the final system_report.txt output.


Sep 27 05:26
ironman@ironman: ~/Desktop
28K College/CSE
12K College/ME/Sem2
12K College/ME/Sem1
12K College/ECE/Sem2
12K College/ECE/Sem1
12K College/Civil/Sem2
Linux ironman 7.0.0-30-generic #30-Ubuntu SMP PREEMPT_DYNAMIC Fri Jul 31 18:22:54 UTC 2026 x86_64 GNU/Linux
ironman
541 tail -n 10 system_report.txt
542 clea
543 clear
544 uname -a >> system_report.txt
545 tail -n 1 system_report.txt
546 clear
547 hostname >> system_report.txt
548 tail -n 1 system_report.txt
549 clear
550 history | tail -10 >> system_report.txt
total used free shared buff/cache available
Mem: 7.6Gi 1.5Gi 4.7Gi 38Mi 1.6Gi 6.0Gi
Swap: 0B 0B 0B
UID PID PPID C STIME TTY TIME CMD
root 1 0 0 02:28 ? 00:00:18 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
root 2 0 0 02:28 ? 00:00:00 [kthreadd]
root 3 2 0 02:28 ? 00:00:00 [pool_workqueue_release]
root 4 2 0 02:28 ? 00:00:00 [kworker/R-rcu_gp]
root 5 2 0 02:28 ? 00:00:00 [kworker/R-sync_wq]
root 6 2 0 02:28 ? 00:00:00 [kworker/R-kvfree_rcu_reclaim]
root 7 2 0 02:28 ? 00:00:00 [kworker/R-slub_flushwq]
root 8 2 0 02:28 ? 00:00:00 [kworker/R-netns]
root 12 2 0 02:28 ? 00:00:00 [kworker/u36:0-ipv6_addrconf]
ironman@ironman:~/Desktop$